

JOSUE TAPIA HERNÁNDEZ

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EDUCATION

Tecnológico de Monterrey

BS in Mathematics and Data Science

Guadalajara, Mexico

Expected Jun 2027

- GPA: 97/100 — Academic Scholarship (Top 5%). Coursework: Data Structures & Algorithms, Stochastic Processes, Statistical Inference, Deep Learning, Optimization.

EXPERIENCE

Co-Founder & ML Engineer

JATA — AgTech Startup

Sep 2025 – Present

Guadalajara, Mexico

- Co-founded an AgTech startup (**1st Place**, UST Hackathon) and shipped an **end-to-end** crop yield forecasting platform for a commercial greenhouse client — **2 greenhouses**, **5 growing seasons**, **365 weekly production records**.
- Built a **CNN-RNN model** in PyTorch (3 residual conv blocks + 3-layer LSTM with a custom correlation-weighted MSE loss) that predicts weekly harvest **4 weeks ahead** with **0.70 R-squared** and **10.7% MAPE** on a held-out test season, ingesting **16 sensor features** (temperature, humidity, CO2, solar radiation, vapor pressure deficit).
- Owned the full data pipeline end-to-end: **357 days** of daily sensor data (**zero missing**), weekly aggregation, PCA reduction, **20-seed robustness sweep**. Now designing a **YOLOv8** computer vision module for automated tomato counting and ripeness classification.

Machine Learning Researcher (Undergraduate)

Tecnológico de Monterrey — Advisor: Prof. Andrés Ramírez Morales

Dec 2025 – Present

Guadalajara, Mexico

- Investigating the implementation of **Graph Neural Networks** (GNNs), specifically **Lorentz Equivariant** architectures, to analyze particle jet symmetries.
- Building a validation framework to reproduce SOTA benchmarks (e.g., LorentzNet), processing **high-dimensional Big Data** with the goal of validating literature performance (**AUC greater than 0.98**).
- Studying asymptotic statistical methods (Cowan) to design a future **inference pipeline** capable of estimating discovery significance and systematic uncertainties.

PROJECTS

DeepProver — LLM Reasoning via Reinforcement Learning | PyTorch, PEFT/LoRA, LeanDojo, FastAPI, React Mar 2026

- Fine-tuned a **7B-parameter LLM** using GRPO + LoRA, building a **custom RL training loop** to support a dual reward signal unsupported by standard RLHF libraries — LoRA rank 16 across **7 modules**, **8 candidate outputs** per input, PPO clipping, KL regularization, bfloat16 + gradient checkpointing.
- Designed a custom RL environment with a **dual reward system**: binary outcome signal + fine-grained per-step process rewards with first-error propagation, enabling the model to learn from **partial successes**. Built a robust output parser handling **35+ domain-specific patterns** with structured block grouping and timeout guards.
- Shipped a **FastAPI + React** interface where users submit problems and watch the model generate and verify solutions live. Developed a **Topological Generation Graph** module that builds edges between proof steps via **Jaccard similarity** and performs **cross-generation tactic credit assignment** to improve the RL reward signal. Open-source with 4-case test suite.

Hive — Autonomous MLOps Agent Platform | LangGraph, Gemini 2.5 Pro, FastAPI, React, Vast.ai

2026

- Architected a **12-agent LangGraph pipeline** that autonomously provisions cloud GPUs, generates training scripts via LLM, and optimizes ML training end-to-end — reducing manual MLOps workflow from **~2 hours to under 10 minutes**.
- Built intelligent GPU marketplace selection using Gemini 2.5 Pro, consistently identifying optimal instances at **\$0.029/hr vs. \$3.50/hr average** (99% cost reduction). Migrated entire remote execution layer to **SSH-based interaction** with Vast.ai across **9+ geographic regions**.
- Shipped a **React + FastAPI real-time dashboard** with live agent activity feed, pipeline state streaming via graph.stream(), and **WebSocket monitoring across 7 API routes**. Built fault-tolerant lifecycle management with automated provisioning, health checks, and deterministic teardown preventing orphaned billing.

AWARDS

ICPC National Regional Contest — Honourable Mention (2x) — Top 17% Nationally

2024, 2025

National Physics Olympiad — Bronze Medalist (Top 20%) — National Training for IPHO

Nov 2022

National Olympiad in Informatics — Bronze Medalist (Top 15%) — National Training for IOI

Nov 2022

National Math Olympiad — Honourable Mention (2x) — Solution officially cited by UADY

2021, 2022

SKILLS

Languages & Core: Python, C++, R, SQL, Bash, Git, Linux/Unix, Lean 4, Data Structures.

AI & ML: PyTorch, TensorFlow, PEFT/LoRA, LeanDojo, scikit-learn, XGBoost, YOLOv8, LangGraph, Gemini API, DSPy, SciPy, pandas, NumPy.

Infra & Web: FastAPI, React, WebSockets, Docker, SSH/Vast.ai, REST APIs, GitHub Actions, Hugging Face Hub.

Concepts: Reinforcement Learning (GRPO, PPO), Deep Learning, Statistical Inference, Computer Vision, Graph Neural Networks, Multi-Agent Systems.

Languages: Spanish (Native), English (C1 Advanced).